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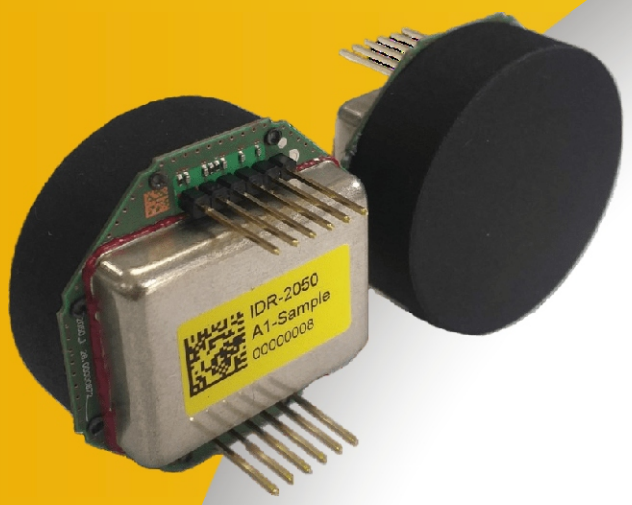
IDR-2050 RADAR SENSOR



The IDR-2050 is a radar sensor operating at 60 GHz. The inte-grated µC unit provides customers an easily integrable sensor for level measurement and other industrial applications. The sensor allows precise distance measurement to stationary and moving objects within a narrow beam. The small, flat design allows integration even into off-the-shelf housings.

With an innovative DSP algorithm and advanced calibration techniques the sensor is able to reach millimeter accuracy.

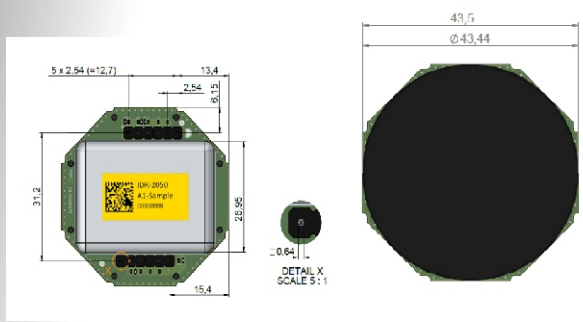
Features

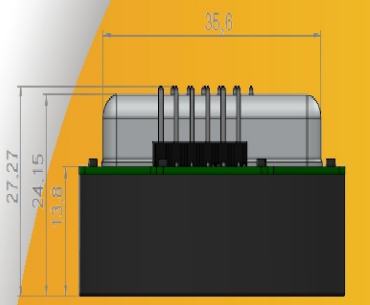
- Radar-based distance measurement sensor operating at 60 GHz
- Distance measurement with millimeter accuracy
- Output of distance, velocity, direction and signal strength of a target via UART
- Small form factor for easy integration into customer housing
- Configurable detection range
- Narrow beam width (azimuth and Elevation)

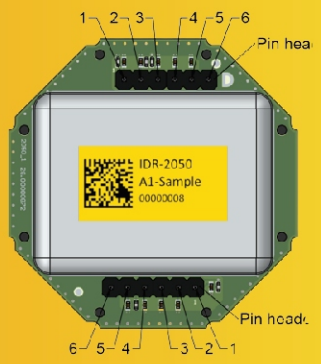
60 GHz Distance Measurement Sensor
Level Measurement
Industrial Applications

- Movement
- Velocity
- Direction
- Presence
- Angle
- Distance

Mechanical Outlines







+91 7017880914
kuchhalbrothers@gmail.com
www.sensormart.in

Specifications

PARAMETER	CONDITIONS	SYMBOL	MIN	TYP	MAX	UNITS
Radar						
Transmit frequencies		f_t	61		63.5	GHz
Output power (EIRP)	ETSI regulation mode	P_{out}		30	30	dBm
	FCC regulation mode			10	10	dBm
Sensor						
Detection distance	ETSI mode ,smallest target : powder	d_r	0		10	m
Accuracy	Depending on the surface of the illuminated object, starting at 0.25m range. Measurements closer typically have an accuracy of ±5cm. Over the suitable temperature range,the measured value varies by ±1cm.			±5		mm
Start up time	Sensor powered up, measurement performed; Fast measurement mode			190		msec
Power supply						
Supply voltage		V_{cc}	4	12	16	V
Average current	@ 5V, Normal Mode, ESSI regulation mode	I_{cc}		379(3)		mA
Environment						
Operating temperature		T_{OP}	-40		+85	°C
Storage dimensions		T_{STG}	-40		+85	°C
Mechanical Outlines						
Outlines dimensions		Height Length Width		27.3 43.5 43.5		mm
Weight						
Sensor weight				26		g

1104,1105, Salempur Rajputana,industrial Area, Roorkee-247667, Uttarakhand